



- PANDAS
- SEABORN
- PLOTLY
- PRATICS
- DATA INSIGHTS

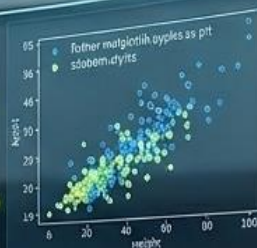
STARTING SOON

```
import matplotlib.pyplot as plt  
  
# Importing data  
data = pd.read_csv('data.csv')  
plt.scatter(data['x'], data['y'])  
plt.show()
```

```
# Data  
plt = matplotlib.pyplot  
plt = matplotlib.pyplot  
plt.show()
```

```
# Seaborn  
plt = sns.relplot(  
    data=data,  
    x='x',  
    y='y',  
    hue='category',  
    style='category',  
    facet_kws={'col': 'category'}  
)  
plt.show()
```

DATA VISUALIZATION



Data Visualization Camp using Python

22 – 25 June 2026



1:30 – 3:00 PM • 90 minutes

Guided Lab

Class begins shortly — please take a seat

STARTING SOON



Data Visualization
Camp using Python

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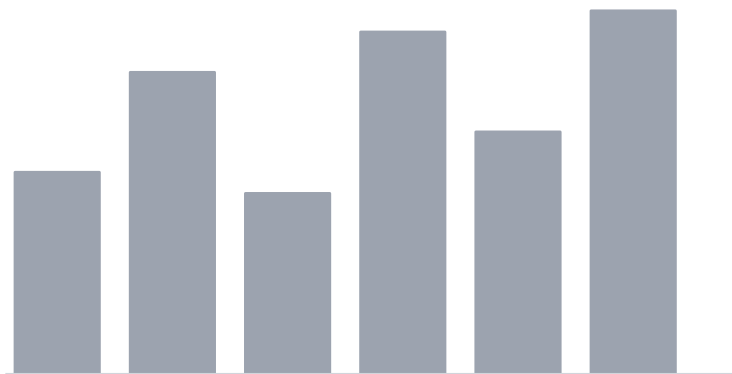


The Challenge: Rescue the Boring Chart!

📖 This exercise is inspired by Murach's Python for Data Science — a book used in college courses!



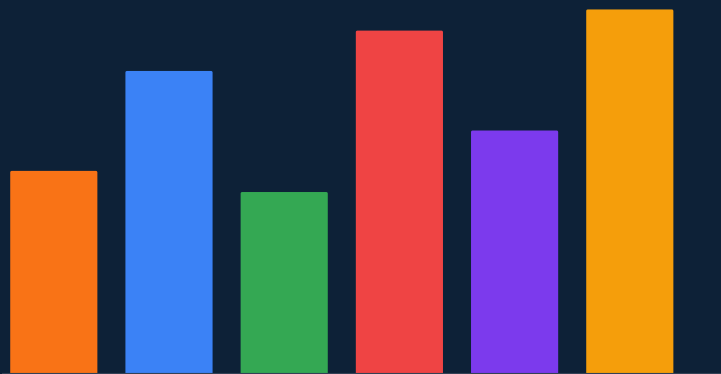
The BORING Version



grey · no labels · no title · no story



YOUR Transformed Version!



colors · labels · title · YOUR story!

Start Here: The Boring Chart (copy this into Colab)

```
import matplotlib.pyplot as plt

# — Murach Exercise: Favorite Sports Survey —
sports = ['Soccer', 'Basketball', 'Tennis', 'Baseball', 'Football', 'Volleyball']
votes = [45, 38, 22, 30, 50, 15]

# This is the boring version 🙄
plt.bar(sports, votes)
plt.show()
```

✗ No title — what is this chart about?

✗ No axis labels — what do the numbers mean?

✗ Gray bars — no visual appeal

Weapon 1: Colors — Make It POP!

```
# Option 1: One color for all bars
plt.bar(sports, votes,
        color='steelblue')
```

```
# Option 2: List of colors
my_colors = ['#FF6B35',
             '#4ECDC4',
             '#45B7D1',
             '#FEE154',
             '#7BC67E',
             '#C19EE0']
```

```
plt.bar(sports, votes,
        color=my_colors)
```

```
# Option 3: Built-in colormap
import matplotlib.cm as cm
plt.bar(sports, votes,
        color=cm.Set3.colors[:6])
```

Try These Color Combos!

Vibrant Pop



Sunset Warm



Cool Ocean



Forest Fresh



Pro tip: complementary colors (opposites on color wheel) look great together!

Weapon 2: Markers & Styles — Add Personality

```
# For bar charts:
plt.bar(sports, votes,
        color=my_colors,
        edgecolor='black',
        linewidth=1.5)

# For line charts:
plt.plot(x, y,
         color='coral',
         marker='o',
         markersize=10,
         linestyle='--',
         linewidth=3)

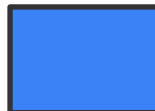
# Rotate x-axis labels
plt.xticks(rotation=45)
```

Visual Differences



No edge

Default — plain



Black edge

edgecolor='black'



White edge

edgecolor='white'



Thick edge

linewidth=3

Markers (for line charts):



Weapon 3: Titles, Labels & Annotations

Make the chart look GREAT:

```
plt.title('Favorite Sports at  
Camp 2026!',  
         fontsize=16, fontweight='bold',  
         color='navy', pad=12)
```

```
plt.xlabel('Sport',  
          fontsize=12, fontweight='bold')
```

```
plt.ylabel('Number of Votes',  
          fontsize=12, fontweight='bold')
```

Add a text annotation to chart:

```
plt.annotate('Most Popular!',  
            xy=(4, 50), xytext=(3.2, 42))
```

```
plt.tight_layout()  
plt.show()
```

What Each Parameter Does

fontsize=16

→ Make the text bigger / smaller

fontweight='bold'

→ Bold the text for emphasis

color='navy'

→ Set the text color

pad=12

→ Add spacing around title

plt.tight_layout()

→ Fix overlapping — always add this!

plt.annotate(...)

→ Add notes on the chart

The Complete Transformed Code

```
import matplotlib.pyplot as plt

sports = ['Soccer', 'Basketball', 'Tennis', 'Baseball', 'Football', 'Volleyball']
votes = [45, 38, 22, 30, 50, 15]
colors = ['#FF6B35', '#4ECDC4', '#45B7D1', '#FEE154', '#7BC67E', '#C19EE0']

# Draw the chart
bars = plt.bar(sports, votes, color=colors, edgecolor='white', linewidth=1.5)

# Add titles and labels
plt.title('Favorite Sports at Camp 2026!',
          fontsize=16, fontweight='bold', color='navy', pad=12)
plt.xlabel('Sport', fontsize=12, fontweight='bold')
plt.ylabel('Number of Votes', fontsize=12, fontweight='bold')
plt.grid(axis='y', alpha=0.3)

plt.tight_layout()
plt.show()
```

Copy this into Colab → Run it → Then customize it YOUR way!

Team Challenge: Make It YOURS!

Day 1

Form teams of 2-3. Every team uses the same data but creates a TOTALLY DIFFERENT looking chart!

Level 1

-  Change all 6 bar colors
-  Add a title with your team name
-  Label both axes
-  Add `plt.grid(axis='y')`

Level 2

-  Add vote counts on top of each bar
-  Rotate x labels 45°
-  Change the chart background color
-  Use `plt.xticks(rotation=45)`

Level 3

-  Add an arrow annotation on the tallest bar
-  Use your own custom color palette
-  Add a horizontal line at the average
-  Save as image:
`plt.savefig('sports.png')`

30 minutes to build! Then Gallery Walk — vote on your FAVORITE chart!

Bonus Challenge: Your OWN Data!

Day 1

Finished early? Pick one of these topics and create a chart using your OWN made-up (or real) data!



Gaming

Hours played per game this month

['Minecraft', 'Roblox', 'Fortnite', 'Valorant']



Music

Top 5 most-played songs (streams)

['Song A', 'Song B', 'Song C', 'Song D', 'Song E']



Food

Your ranking of school lunch options

['Pizza', 'Burger', 'Tacos', 'Salad', 'Pasta']



Social

Minutes per day on each app

['TikTok', 'Instagram', 'YouTube', 'Snap', 'Discord']



Weather

High temperature each day this week

['Mon', 'Tue', 'Wed', 'Thu', 'Fri', 'Sat', 'Sun']



Budget

How you'd spend \$100 on entertainment

['Movies', 'Games', 'Food', 'Clothes', 'Other']

Gallery Walk + Day 1 Wrap-Up

Day 1

Gallery Walk Rules

Each team displays their chart

Use sticky notes to give feedback:

- What you loved
- One suggestion

Pro Design Tips

- ✓ Every chart MUST have a title & axis labels
- ✓ Max 7 colors in one chart
- ✓ High contrast = easy to read
- ✓ Less is more
- ✓ The most important data should pop most
- ✓ Always ask: 'What story does this tell?'

Day 1 Accomplishments:

Python setup in Google Colab

Matplotlib: plot, scatter, bar, labels

Turned a boring gray chart into something beautiful!

Colors, markers, and design principles

Tomorrow: Day 2 — Pandas, Real Data Files & The Mystery Dataset Challenge!

See You Tomorrow!

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